

Radar-Vision Fusion Sensor ET-RC8083 Datasheet

- Product Type: Radar-Vision Fusion Sensor
- Model: ET-RC8083
- Doc Type: Datasheet
- Doc Version: v1.0
- Doc Release Data: March 25,2026

1. Product Overview

ET-RC8083 is an all-in-one radar-vision fusion sensor, which integrates millimeter-wave radar and high-definition camera to realize **all-weather, full-element, high-precision** traffic perception. It supports vehicle flow, speed, queue length, trajectory tracking, traffic event detection, license plate recognition and vehicle classification, with high perception accuracy.

2. Key Features

- Radar + vision fusion technology for high accuracy and stability
- 80GHz wide-area radar for long-distance detection
- 2K high-definition camera with night vision
- All-weather operation (rain, fog, strong light, darkness)
- Real-time traffic data output and analysis
- Wide voltage input and rugged IP67 enclosure
- Stable performance in harsh outdoor environments

3. Typical Applications

- Traffic flow statistics and analysis
- Intelligent traffic signal control
- Intersection vehicle approach warning
- Roadside illegal parking monitoring
- Congestion and queue length detection
- Vehicle classification and behavior analysis
- Perimeter security monitoring

4. Technical Specifications

General

- Model: ET-RC8083 (Radar and vision fusion sensor)
- Working Voltage: 9~36V DC
- Power Consumption: Approx. 3W
- Operating Temperature: -40°C ~ +85°C
- Humidity: 0-95% RH (non-condensing)
- Protection Grade: IP67
- Interfaces: RJ45 Ethernet, RS485, CAN2.0B

Radar Parameters

- Frequency Band: 79-81 GHz
- Detection Range: Up to 350 m
- Distance Accuracy: ± 0.2 m

- Distance Resolution: 1.4 m
- Speed Range: 0-337 km/h
- Speed Accuracy: ± 0.05 km/h
- Azimuth Coverage: 8 lanes
- Azimuth Accuracy: $\pm 0.5^\circ$ @ 0°
- Targets: 200+
- Refresh Rate: 10 Hz

Camera Parameters**Camera 1 – Traffic Monitoring**

- Resolution: 2560×1440 (2K)
- Sensor: 1/3 inch
- Focal Length: 16 mm
- Frame Rate: 30 FPS
- Function: Traffic flow, vehicle classification, night vision

Camera 2 – License Plate Recognition

- Resolution: 2560×1440 (2K)
- Sensor: 1/3 inch
- Focal Length: 25 mm
- Frame Rate: 30 FPS
- Function: License plate recognition (LPR)

5. Core Functions

- Lane-level traffic volume statistics
- Real-time speed & average speed calculation
- Queue length measurement & congestion detection
- Time headway / space headway analysis
- Space occupancy & time occupancy rate
- Vehicle classification (car, truck, motorcycle, non-motor vehicle)
- License plate recognition (LPR)
- Illegal parking, wrong-way driving & incident detection
- Night vision & all-weather detection

5. Appearance

(Images shown for reference; actual configuration subject to delivery)

Dimension: 160mm × 120mm × 51.5mm

